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Rossini Martyr

Data Scientist

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Technical Skills

Languages **Proficient**: Python, SQL
Experience: Pytorch, TensorFlow

Tech Stack Git, Streamlit, AWS Sagemaker, Docker, RAG, Spyder, FastAPI, Langchain, LaTeX, Tableau

Experience

Senior Data Scientist/Engineer **Feb 2024 — Present**
Fast + Epp *Toronto, Canada*

- Managed a team of engineers on projects with assets valued over $250M, framing Owner requirements into analytical problems with clear goals and constraints while ensuring projects are delivered on time and on budget
- Evaluated design alternatives with Consultant team, explaining tradeoffs of material choice and aesthetics to the Owner in order to facilitate informed decision making, yielding $200,000+ in direct capital cost savings
- Reviewed Python code for embodied carbon calculator, updating underlying databases and adding new UI elements to improve user experience, simplifying reporting to meet industry carbon reporting commitment

Senior Data Scientist/Engineer **Apr 2016 — Feb 2024**
Arup *Toronto, Canada*

- Developed Python and SQL scripts to automatically parse and manipulate millions of rows of time series data from Amazon Redshift databases, providing instantaneous results and reducing manual data wrangling time by over 50%
- Automated the creation of structural analysis models using Python, reducing analysis time by over 30% by automatically connecting to internal software via API's and eliminating potential errors due to manual data processing
- Assessed seismic hazard risk for building renovations, evaluating likely capital expenditure against probability of catastrophic loss, delivering over $100,000 in cost savings through seismic risk modelling

Research Scientist **Sep 2013 — Apr 2016**
University of Toronto *Toronto, Canada*

- Calibrated a predictive model using large-scale experimental data to estimate the strength of structural steel components under earthquake loads, improving accuracy by over 30% compared to conventional methods
- Defined a cumulative probability distribution to estimate the likelihood of component failure, validating model parameters with physical testing and establishing clear confidence bounds for all predictions
- Programmed automation routines using FORTRAN and Excel, reducing data analysis and visualization time by 20%

Projects

Using ML to Estimate Catastrophe Loss due to Flooding (Link) **Jan 2026 — Feb 2026**

- Built an XGBoost model which used asset-specific characteristics to estimate flood damage costs based on historical insurance claims, improving accuracy by 33% compared to the baseline. The model was deployed using Streamlit, allowing an underwriter to quickly estimate financial loss and to categorize riskiness of new assets

Education

Master of Applied Science (GPA 3.94/4.00), *University of Toronto* June 2016
B.A.Sc in Civil Engineering (GPA 3.88/4.00), *University of Toronto, (Graduated #1 in cohort)* June 2013